

A PROTOTYPE PLC BUILT AUTOMATIC VEHICLE WASHING SYSTEM USING VFD

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ABSTRACT

Sustained infrastructural development in this modern era leaves vehicle surface exposed daily to grime. Although manual vehicle washing has been in existence since vehicle invention, it is arduous, time consuming and uses water excessively. This is not to mention petite abrasion and rapid loss of vehicle surface brilliance with manual vehicle washing. Automatic vehicle washing industry has improved from its first innovation in the twilight of 1930's through advancement in technology, the system is more reliable and efficient. This paper's emphasis is on the development of a scalable prototype exterior automatic vehicle washing system. The system is built around a Logo Zelio (SR3B261BD) PLC for inputs and output components control. VFD moderates the conveyor belt motor speed as required for its smooth operation. Water recirculation system is incorporated for water and detergent economy. The status lamp on the control panel informs the user of the instantaneous stages of operation and control of the entire system. Experimentation of the system presents an efficient washing process between work station and minimal water and detergent usage with the incorporated recirculation system. Application of control panel for monitoring and control enhance the cost effectiveness of this system compared to its SCADA counterpart. In the future, the authors will like to integrate dust particle detection sensor for improved surface cleaning.

Keywords: automatic vehicle washing system, PLC, sensor, ladder logic

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1. INTRODUCTION

Sustained infrastructural development in this modern era leaves vehicle surface exposed daily to grime. Although manual vehicle washing has been in existence since vehicle invention, it is arduous, time consuming and uses water excessively (May et al., 2019). This is not to mention petite abrasion and rapid loss of vehicle surface brilliance with manual vehicle washing (Gaikwad et al., 2017). Automation relieves humans of physical contact and sensory

input toward the target process (Sorkhabi and Khazini, 2013). Hence, the advancement in technology has improved significantly the reliability and efficiency of vehicle washing through automation of the process. Automatic vehicle washing industry has improved from its first innovation in the twilight of 1930's (Meena et al., 2017) to the programmable logic controller (PLC) vehicle washing system in the twenty first century. PLCs are a unique class of microprocessor-built controller integrated with programmable memory for instruction storage with logical, sequential, timing, counting and arithmetic function implementation (Bolton, 2006). The outcome is the control of machines via pre-stored program. Automatic vehicle washing system work station comprise of the; showery down, foaming, washing, rinsing and drying. In this project, the authors adopted the cyclic process presented by Rana et al (2017) for its 90 percent cleaning aftermath. Automatic vehicle washing system are complex and expensive system; since they combine PLC and supervisory control and data acquisition (SCADA). Hence, they are found in developed countries and in public locations. The objective of this paper however, is to develop a scalable prototype automatic vehicle washing system. Affordable for industrial application even in developing countries and private application by minimizing the component list for cost reduction. This system is integrated with water recirculation system for water economy and environmental protection. Also the authors replaced the SCADA with control panel which comprise of status lamps; this informs the user of the instantaneous stages of operation and control of the whole system.

1.1. Related works

Shaikh et al (2019); Lalluwadia et al (2017); Gaikwad et al (2017) presented an automatic vehicle washing system controlled by PLC with SCADA. The system is based on the cyclic process for effective cleaning. However, water recirculation was not incorporated; thus leading to excessive water usage. Although the system is effectual, the combination of PLC and SCADA increases the cost of production significantly particularly in developing countries. Sorkhabi and Khazini (2013); Meena et al (2017) proposed an automatic vehicle washing system using PLC. The authors addressed the system cost of production by reducing required hardware components. However, water recirculation was not integrated for water management. May et al (2019) described hardware and software simulation of an automatic vehicle washing system with Mitsubishi (FX 2N-25MR) PLC. System experimentation explained its efficacy. Nevertheless, lack of water recirculation increases water usage and environmental pollution. Also, authors did not present incorporation of SCADA or control panel for system operations remote monitoring. Vidyasagar et al (2015) proposed the concept of automatic vehicle washing with PLC and RFID-GSM platform for operation completion status notification to the user. Authors focus is on hardware components reduction to achieve cost requirement objective. This was done by adapting existing GSM technology for finish operation status communication to the user. Tryouts revealed model practicability, nonetheless, water recirculation with instantaneous operation status notification is equally important for system operation optimization. PIC 16F778A microcontroller was used to control the vehicle washing system. Utekar et al (2015) developed an automatic vehicle washing system with two robotic arms. The arms were replacement of the twin rotary brushes of the conventional automatic vehicle washing system. A Graphical User Interface (GUI) was developed on a PC to control the system. Inputs from the GUI is transmitted via a USB to UART converter to the microcontroller. Water usage was monitored by the vehicle profile detection using an IR sensor and interfacing the output with the water nozzle. Nozzle is kept open while output is available from the IR sensor. Although their model is promising for more effective cleaning, however, the application of robotic arms becomes a crucial cost component. Zhong, et al (2017) introduced the concept of Internet of Things (IoT) and big data to automatic vehicle washing system. The vehicle wash cloud and terminal interface was

created using IoT and big data. This built a synergy of cloud operators, vehicle washing shops, vehicle washer manufacturers and customers. Their system is based on customer feedback on service delivery in using any of their automatic vehicle washing outlets. This platform is based on Linux operating system, which is a stable multi user network operating system. This architecture is divided into 5 layers: data layer, data access layer, services layer, authorization and access control layer and application layer. The data layer uses an open source MySQL database that can be distributed or clustered according to performance requirements. Mhaske et al (2016) used Wecon PLC module for automatic control of a vehicle washing system prototype. PLC was interfaced with the PC via RS-232 correspondence link for downloading or transferring program of the system. The system is compact and adaptable for private applications, but, water recirculation with instantaneous operation status notification is not incorporated. Akilandeswari et al (2014) integrated a PLC with GSM modem in an automatic vehicle washing system. Status information such as vehicle entry/exit and emergency stops is sent to operator and customer via SMS through the GSM-PLC platform. Similarly, the control of the system is achieved through SMS. HORNER PLC was connected to the PC through a RS-232 communication cable for downloading or uploading the program. System hardware is reduced base on PLC-GSM platform application leading to cost reduction. Conversely, GSM is network based system; inadequacy of such services by network providers will affect the practicability of the proposed method. Therefore, development of practicable automatic vehicle washing system with minimal hardware component for marginal cost objective actualization; notwithstanding demands the complement of a water recirculation system with a real-time operation status notification and control. This is the area the authors aim to address in this paper.

2. MATERIALS AND METHODS

2.1. Hardware implementation

Circuit diagram of this system is presented in figure 1. The system is built around Logo Zelio (SR3B261BD) PLC, this is the nucleus of this system. This PLC controls the systems input and output component operations. From figure 1, A 220 AC source voltage is coupled to a Miniature Current Circuit Breaker (MCCB) for effective electrical power control. A power supply module converts the 220 AC source voltage from the MCCB to 24 V DC, because PLC operates on 24 V DC. Digital or analog input and output components are connected to the PLC through the terminal block and address. The transport conveyor is driven with an AC motor; this convey the vehicle through the system work stations. A variable frequency drive (VFD) is used to regulate the speed of the AC motor via it frequency from 60 Hz to 14 Hz. The motor is powered by the 220 V AC, control panel, pump, relays, inductive sensor are powered by the 24 V DC. Control and monitoring of the system's operation is achieved by an operator through the control panel. The panel in figure 2, operator control the system's operation via the green, small red and large red push buttons; to start operation, end operation and emergency stop respectively. Status lamp; green, amber and red on the control panel informs the operation of the system; start, washing and stop instantaneous work stations.

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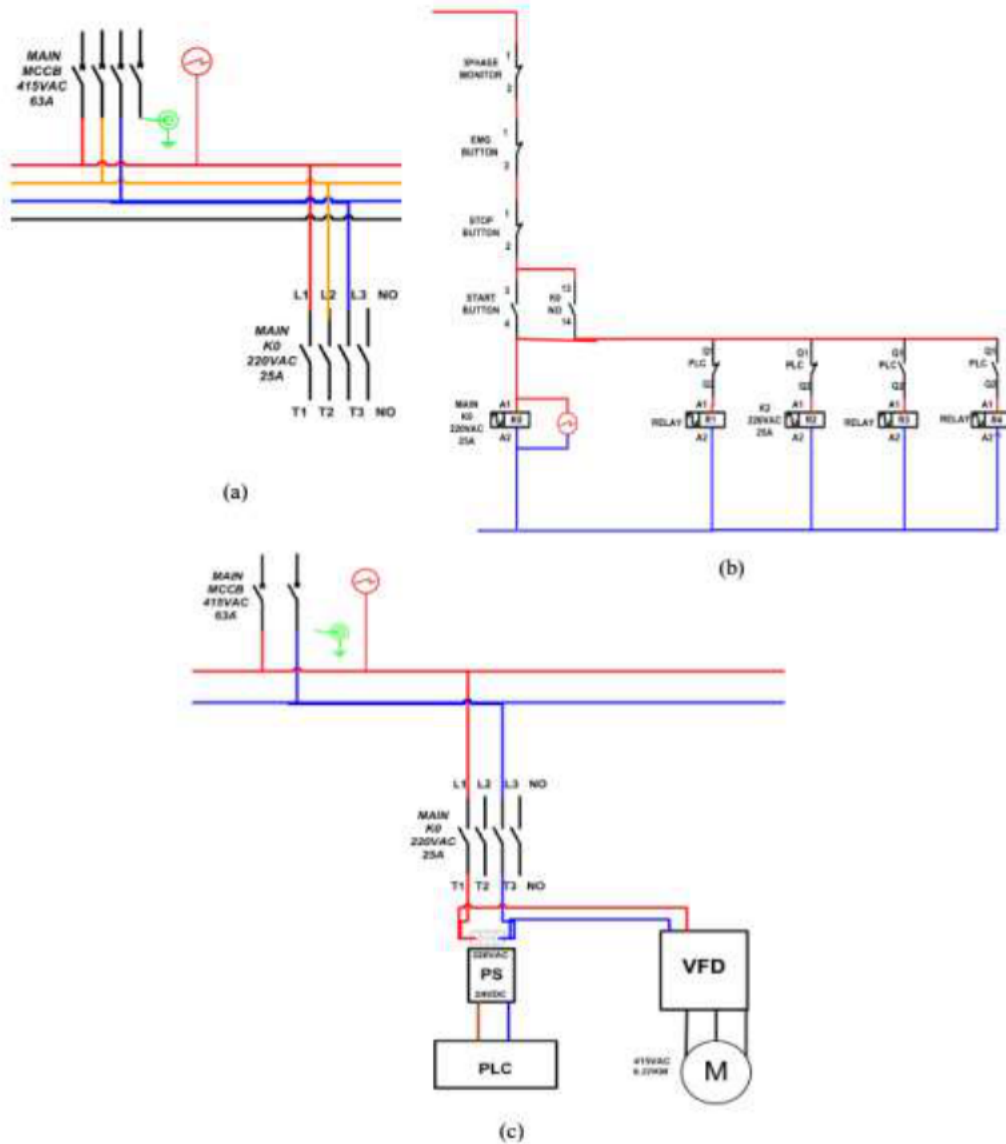


Figure 1 Circuit diagram

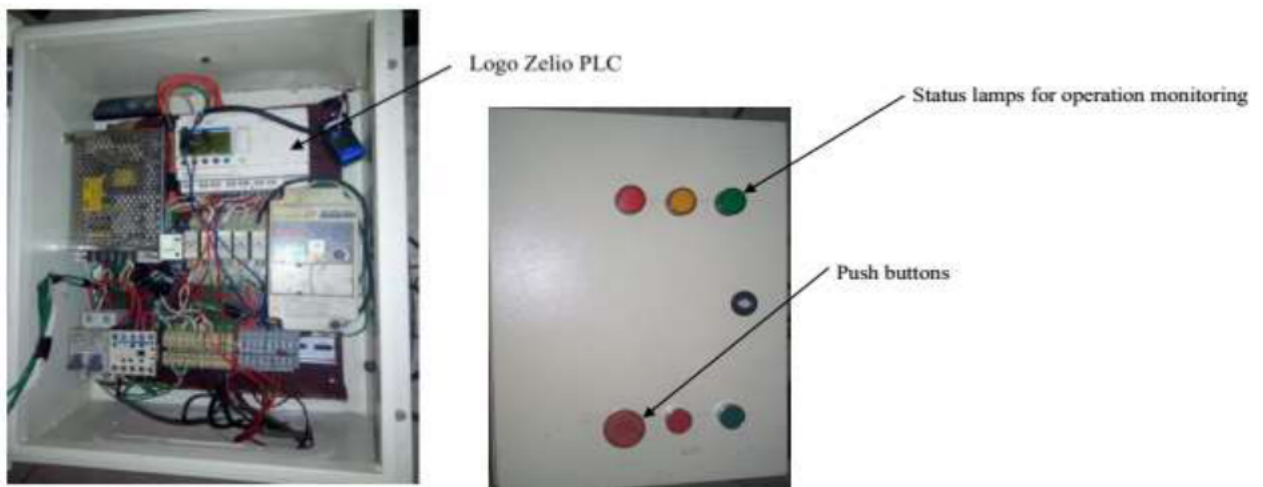


Figure 2 Control panel

2.2. System flowchart

Figure 3 and 4 presents the flowchart of the automatic vehicle washing system showing the sequence of the process and the systems block diagram respectively.

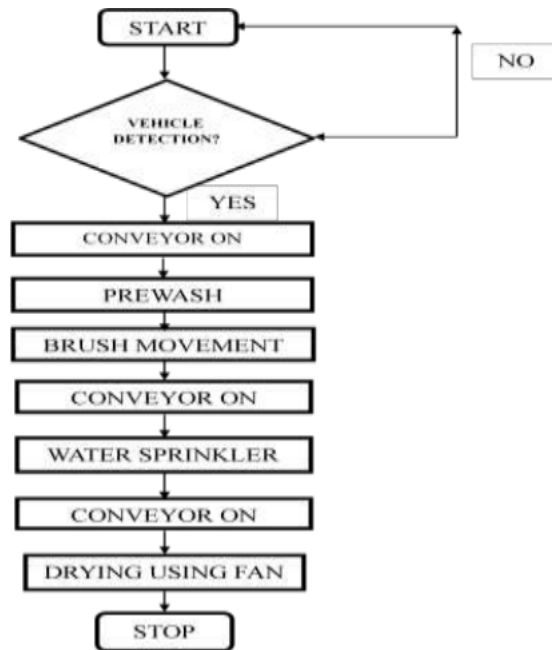


Figure 3 System flowchart

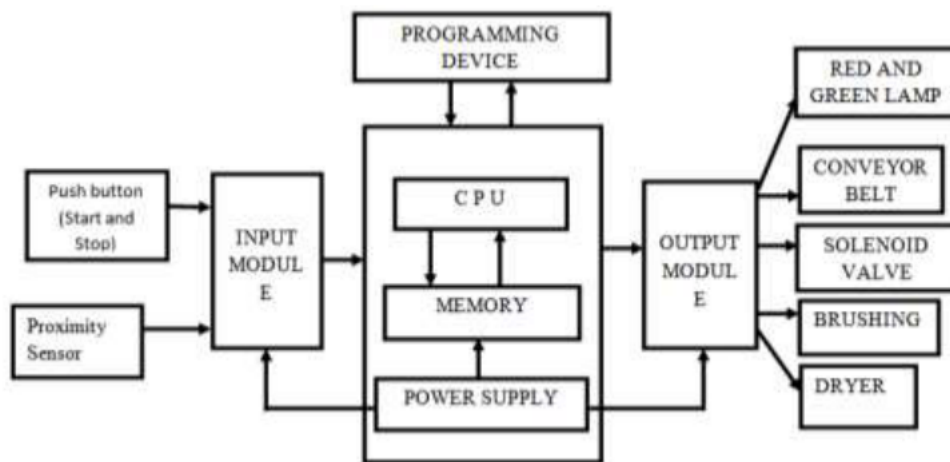


Figure 4 System block diagram

2.3. System operation

With the vehicle parked on the transport conveyor, a capacitive proximity sensor at this entry point detects the presence of the vehicle. This gives an output that energize the AC motor to drive the conveyor leading the vehicle into the washing tunnel. Vehicle reaches the showery down/foaming stage. A proximity sensor at this stage detects the vehicle, energizes a solenoid pump to discharge water and foaming water over the vehicle. This stage spans for 15 sec. The off-delay timer arrangement stops the solenoid pump. A proximity sensor at the washing stage detects the vehicle, energize the twin cotton brush to commence the exterior brushing of the already lather laden vehicle. This continues for a period of 10 sec. Then the off-delay timer deactivates the cotton brushes. Conveyor moves the vehicle to the rinsing stage. Here another proximity sensor detects the vehicle and energizes the solenoid pump to discharge water on

the vehicle for rinsing for 10 sec. the pump is de-energized via the off-delay timing instruction. Conveyor moves the vehicle till it is detection by the drying stage proximity sensor. This detection activates the twin fans on the side of the tunnel for 20 sec. this dry's the vehicle. This fans are deactivated after the specified time by off-delay timing arrangement. The conveyor moves the vehicle to the terminal of the tunnel for finish operation and the cycle is repeated for the succeeding vehicle. Table 1 and 2 below presents the input and output components of the system.

Table 1. Input components

S/N	Input	Address	Tag no
1	Start Push button	I 0.1	Activate the operation
2	Stop push button	I 0.2	Stop the operation
3	Capacitive sensor 1	I 0.3	Capacitive sensor for brushing
4	Capacitive sensor 2	I 0.4	Capacitive sensor for rinsing
5	Capacitive sensor 3	I 0.5	Capacitive sensor for drying

Table 2. Output components

S/N	Output	Address	Tag no
1	Green LED	Q 4.0	Indicate washing process
2	Motor	Q 4.1	Energize the conveyor
3	Yellow LED	Q 4.2	Energize the conveyor light
4	Brush	Q 4.3	Activate the brush
5	Pump 1	Q 4.4	Activate the pump
6	Pump 2	Q 4.5	Energize the rinse water
7	Fan	Q 4.6	Activate the dryer and exit section
8	Pump 3	Q 4.7	Recycling pump

2.4. Water recirculation

Beneath the conveyor is a water recirculation tank. Every liquid discharged on the vehicle drips on the conveyor and flows into the tank underneath. This tank has two compartment separated by a strainer. The used liquids are collected in the first compartment and strained as it flows to the second compartment. This prevents the dirt's passage into the second compartment. The water in the second compartment is lifted with a solenoid pump into the foaming water tank. This reduce the volume of water and detergent required to make lather for succeeding wash. Hence, water and detergent economy is achieved.

2.5. System simulation

Ladder logic is one of the methods of programing a PLC (Bolton, 2006). Therefore, for simplicity, it is adopted in this work. Simulation has been explained to be very crucial for evaluation of a proposed real-world system (Martins et al., 2013). Thus, simulation of the ladder diagram of the Logo Zelio (SR3B261BD) PLC is presented in the subsequent figures.

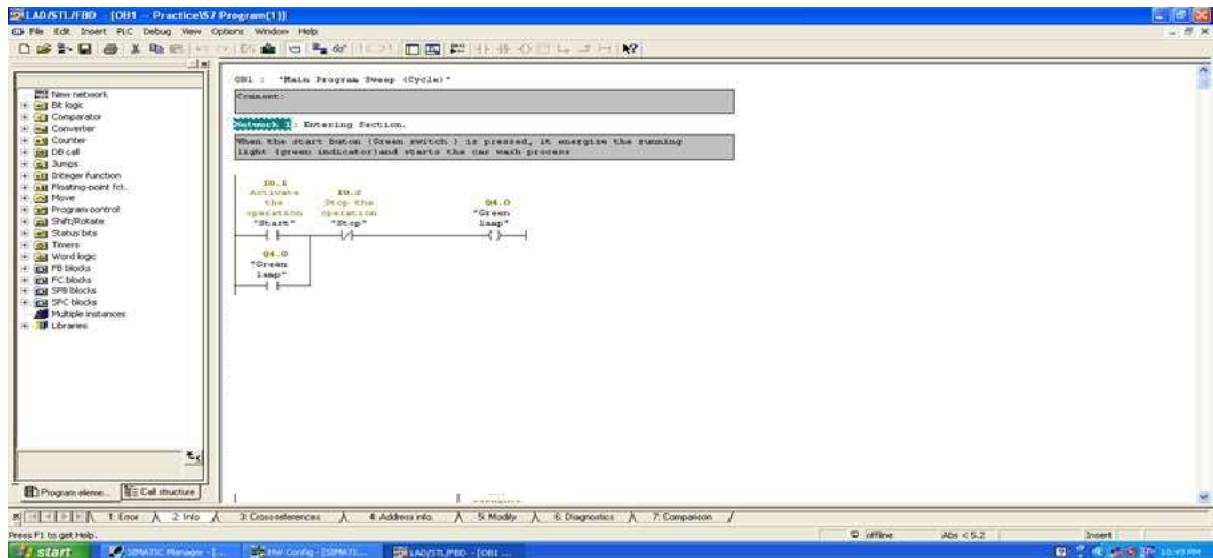


Figure 5 Start simulation

Form figure 5; first rung of the ladder, I0.1 is a normally open contact (NOC), I0.2 is a normally closed contact (NCC). Q4.0 is the output on this rung and it is latched with I0.1. when the push button I0.1 is pressed, it energizes Q4.0 and it stays energized through the latched arrangement.

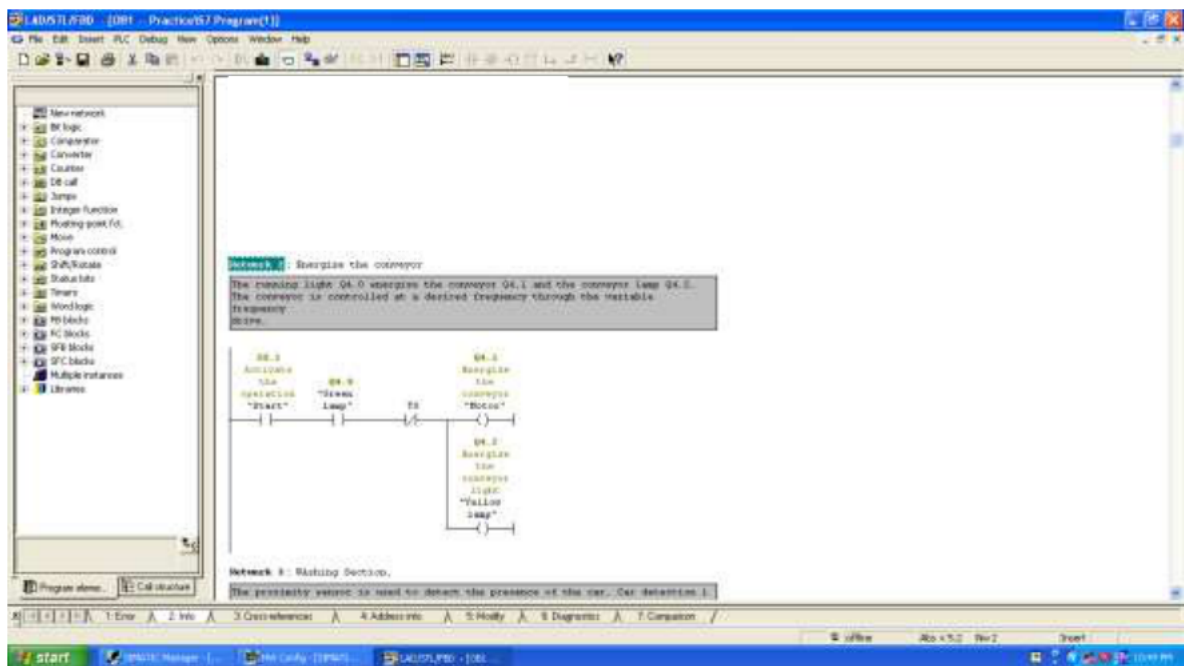


Figure 6 Start conveyor

The second rung in figure 6 shows input I0.1 and Q4.0 as NOC, timer T0 as NCC, with output Q4.1 and Q4.2 latched together. Q4.0 energizes Q4.1 and Q4.2

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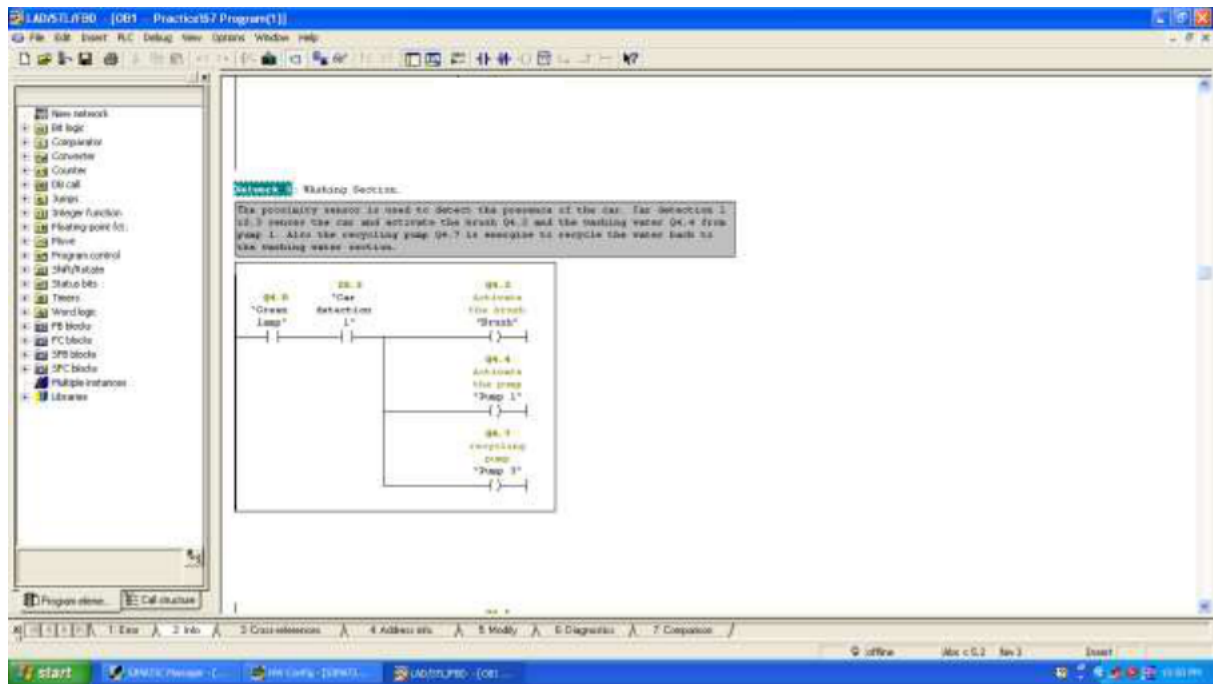


Figure 7 Washing operation

The third rung in figure 7 presents inputs Q4.0 and I0.3 (proximity sensor) as NOC with outputs Q4.3, Q4.4 and Q4.7 latched together to be energized by single input. When I0.3 changes state due vehicle detection all outputs are energized.

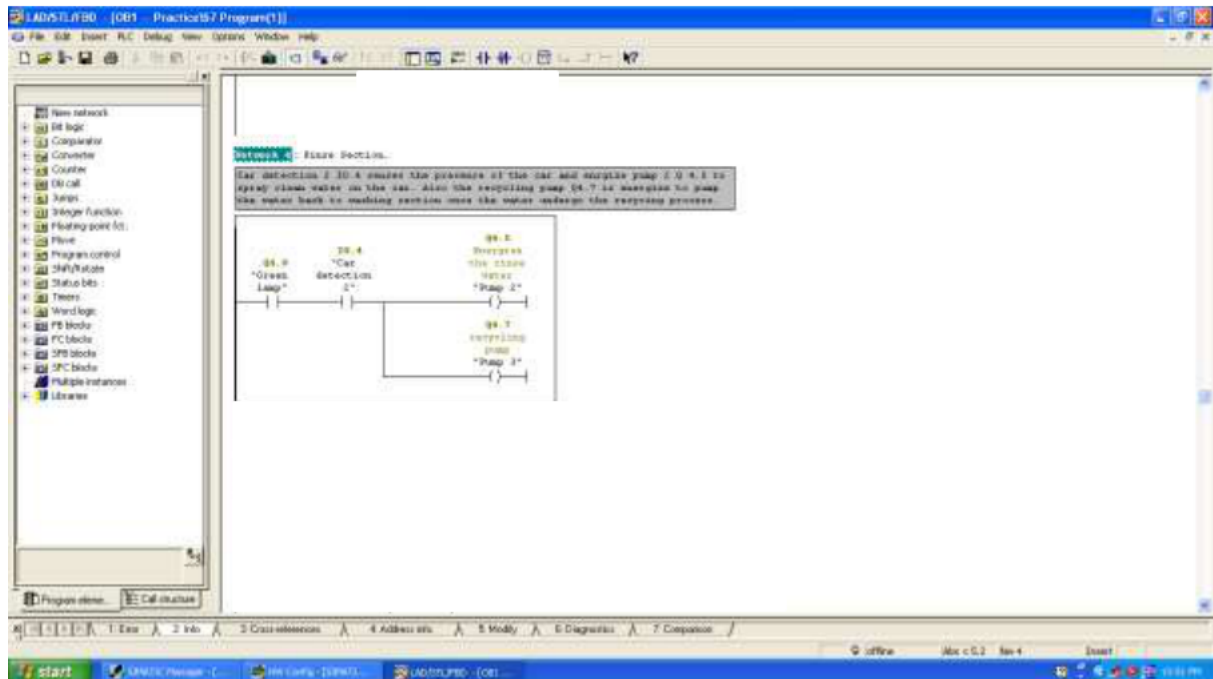


Figure 8 Rinsing stage

The forth rung in figure 8 shows inputs Q4.0 and I0.4 (proximity sensor) as NOC with outputs Q4.5 and Q4.7 latched together. Sensor I0.4 detects the presence of the vehicle in the rinsing stage and changes state. This energizes the outputs on this rung.

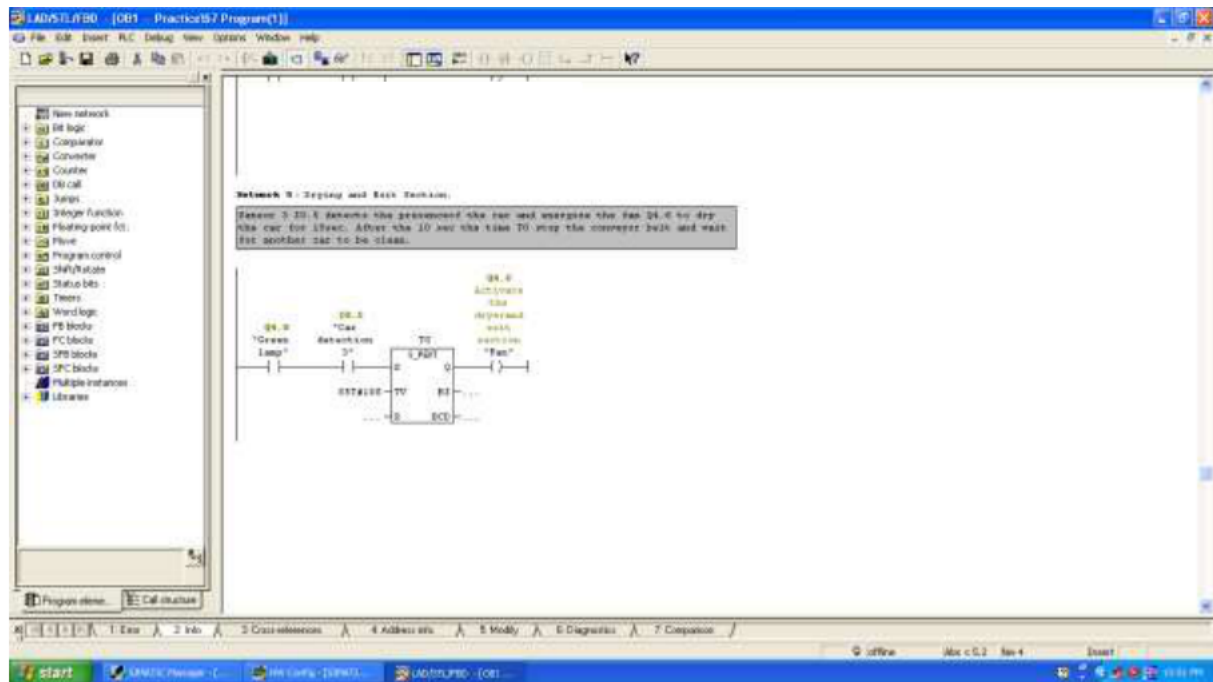


Figure 9 Drying and exit stage

This is the fifth rung, Q4.0 and I0.5 (proximity sensor) are NOC inputs and the timer reset register with Q4.6 as outputs. Vehicle detection by I0.5 in the drying stage changes the state of this contact. This energizes the output accordingly.

Figure 10 shows the implementation of the prototype.

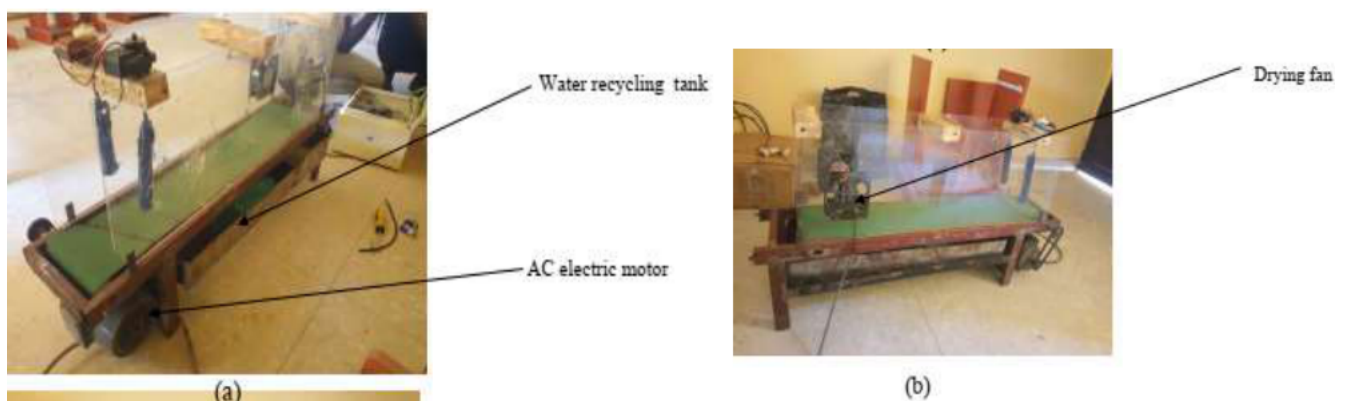


Figure 10 Prototype implementation

3. CONCLUSION

Deduction from the simulation and prototype implementation of the proposed automatic vehicle washing system is that; adoption of control panel with status lamp for operation monitoring and control of the entire system makes this system cost effective in relation to it SCADA counterpart. Also, the integrated water recirculation improve it water and detergent economy, hence, applicable where water economy and soil protection is paramount. These facts will increase the demand and implementation of this design for public and private applications. In this work, authors built the system control around the Logo Zelio

(SR3B261BD) PLC. PLCs are capable of controlling complex system with simple program as presented in our simulation. Compared with other microcontrollers, PLC programming languages are rather simpler to comprehend effectively. Although there are several programming languages for PLC, but in this work we adopted the ladder logic. The simulation with the ladder diagram presents the practicability of the system. Authors use VFD for speed control of the conveyor motor to show the possibility of using AC motor compared to the DC motor found in literature. Meanwhile, the practical implementation might not express the esthetics of the real-world design, however, it presents its viability and scalability. In the future, integration of dust particle detection sensor will enhance the surface cleaning of the system.

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